

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Multimeter Kit, Part Number 3400162, or equivalent
- Multimeter pressure sensor, Part Number 3164491, or equivalent
- Aftertreatment diesel exhaust fluid (DEF) diagnostic line, Part Number 5299833
- Cummins® electronic service tool or equivalent

Additional Service Items

- Female Compuchek™ fitting and mating adapter

General Information

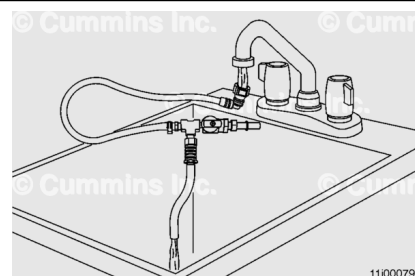
The purpose of this test is to check the ability of the dosing unit to create vacuum.

The measurement is performed using the Aftertreatment Diesel Exhaust Fluid System Leak Test in the recommended Cummins® electronic service tool or equivalent.

Setup Information

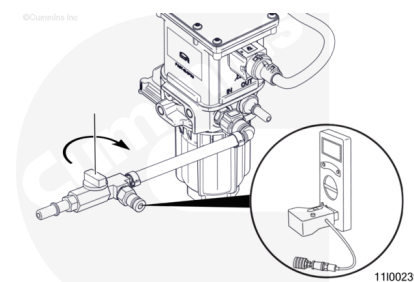
Clean the aftertreatment DEF diagnostic line.

- Verify the DEF is **not** frozen. If the DEF is frozen, run the engine to allow the DEF system to thaw.
- Connect an open female Compuchek™ fitting onto the male Compuchek™ fitting of the diagnostic line.
- Thoroughly flush the diagnostic line. Use warm, clean water. Make sure the Compuchek™ fitting is free of DEF crystal buildup.



- Zero the pressure sensor and connect the DEF diagnostic line.

- Connect the multimeter pressure sensor to the Compuchek™ fitting on the diagnostic line.
- Verify the value at the diagnostic line is open to atmosphere and stationary before the pressure sensor is zeroed.
- Zero the sensor.
- Disconnect the intake line from the DEF dosing unit.
- Disconnect the outlet line from the DEF dosing unit.
- Install the aftertreatment DEF diagnostic line to the intake fitting on the aftertreatment DEF dosing unit.
- Close the valve on the diagnostic line.
- Connect the electronic service tool.



Measure

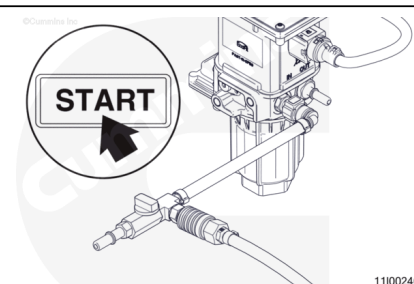
The pressure sensor is sensitive to movement and vibrations. Be careful not to bump the multimeter pressure sensor adaptor or zero could be lost.

Run the Aftertreatment Diesel Exhaust Fluid System Leak Test found under the Engine Control Module (ECM) Diagnostic Test menu in the electronic service tool.

Allow the vacuum reading to stabilize. Usually, this will occur within 15 seconds.

Measure the dosing unit vacuum.

Dosing Unit Vacuum		
mm-hg		in-hg
-203	MIN	-8
-762	MAX	-30



After taking the reading:

- Open the diagnostic line valve.
- Stop the Aftertreatment Diesel Exhaust Fluid System Leak Test.
- Remove the aftertreatment DEF diagnostic line.

If the dosing unit vacuum is **not** within the specification, remove and replace the DEF dosing unit inlet screen filter and perform the DEF dosing unit liquid assisted manual purge.

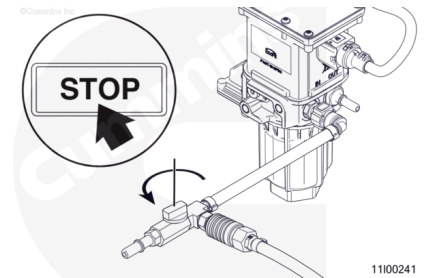
Refer to Procedure 011-143 in Section 11.

(/qs3/pubsys2/xml/en/procedures/377/377-011-143.html)

Finishing Steps

Note : If directed to this procedure from a troubleshooting tree and the test results were within specification, connect the diagnostic line between the dosing unit intake fitting and intake and return to the troubleshooting information.

- Connect the intake line to the DEF dosing unit.



Clean the aftertreatment DEF diagnostic line.

- Connect an open female Compuchek™ fitting onto the male Compuchek™ fitting of the diagnostic line.
- Thoroughly flush the diagnostic line. Use warm, clean water. Make sure the Compuchek™ fitting is free of DEF crystal buildup.

